Paper Review • World Action Model

# WorldBridge: Task Generalization in World Action Models via a “Video→Action Bridge”

WorldBridge: Bridging Video and Action for Task Generalization in World Action Models

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arXiv 2610.40748 • First submitted 2026-10-29

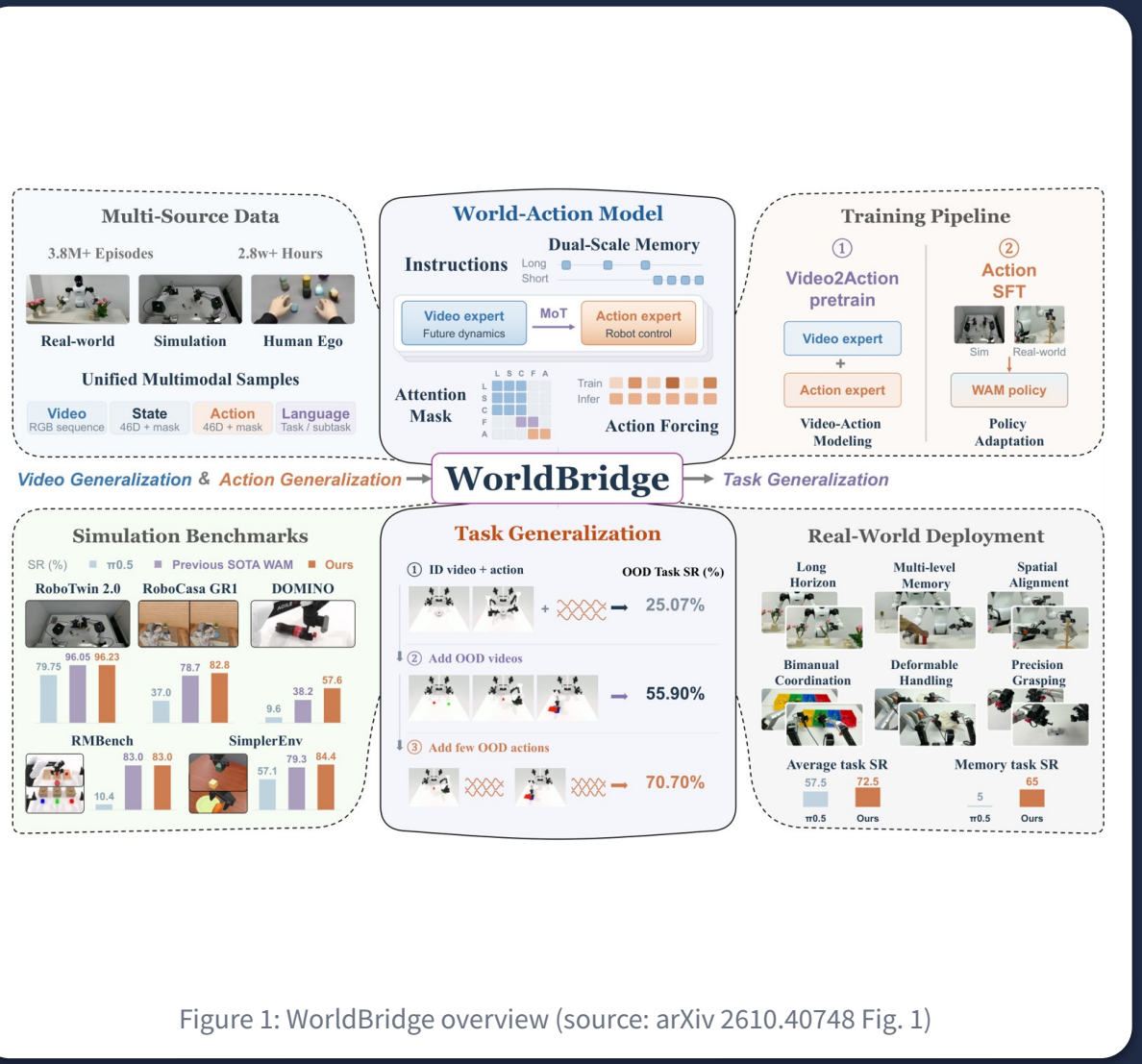



Figure 1: WorldBridge overview (source: arXiv 2610.40748 Fig. 1)

# TL;DR: video widens coverage; a few action demos connect the "bridge"

Decomposes WAM task generalization into **video generalization** (predicting how new tasks unfold visually) and **action generalization** (video→action bridge ≈ inverse dynamics model): unlabeled videos broaden coverage, while very few paired demos calibrate the bridge. The resulting WorldBridge leads on multiple manipulation benchmarks and two real robots.

**25.07→55.90%**

**+ Unlabeled videos**

RoboTwin 2.0 OOD task success +30.83 pts

**→ 70.70%**

**+1 action demo per task**

Another +14.80 pts; 20/task reaches 90.03%

**96.23%**

**RoboTwin 2.0 (full)**

Clean2Random 80.9% , RoboCasa GR1 82.8%

**72.5%**

**Real robot: 6-task avg.**

π0.5: 57.5%; memory task 65% vs 5%



Source: arXiv 2610.40748 abstract, Tables 2/3, Fig. 7

Research Question

# Task generalization = video + action; the bottleneck: video→action bridge

## Video generalization

Predict the visual evolution of tasks beyond the training distribution— “how should the new task unfold?”

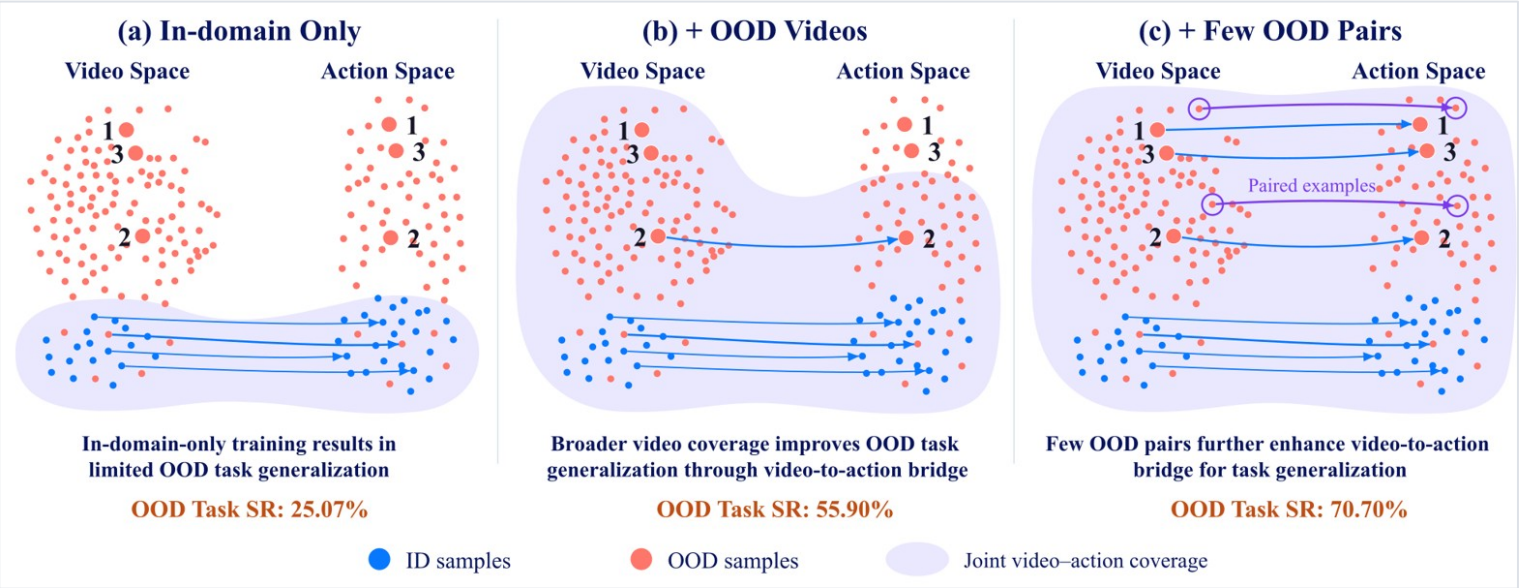
Video offers dense physical-interaction supervision beyond action-label coverage.

## Action generalization

Convert the predicted visual evolution into executable actions.

This step is the video-to-action bridge, acting like an inverse dynamics model (IDM).

A correct video prediction still fails without grounded actions.

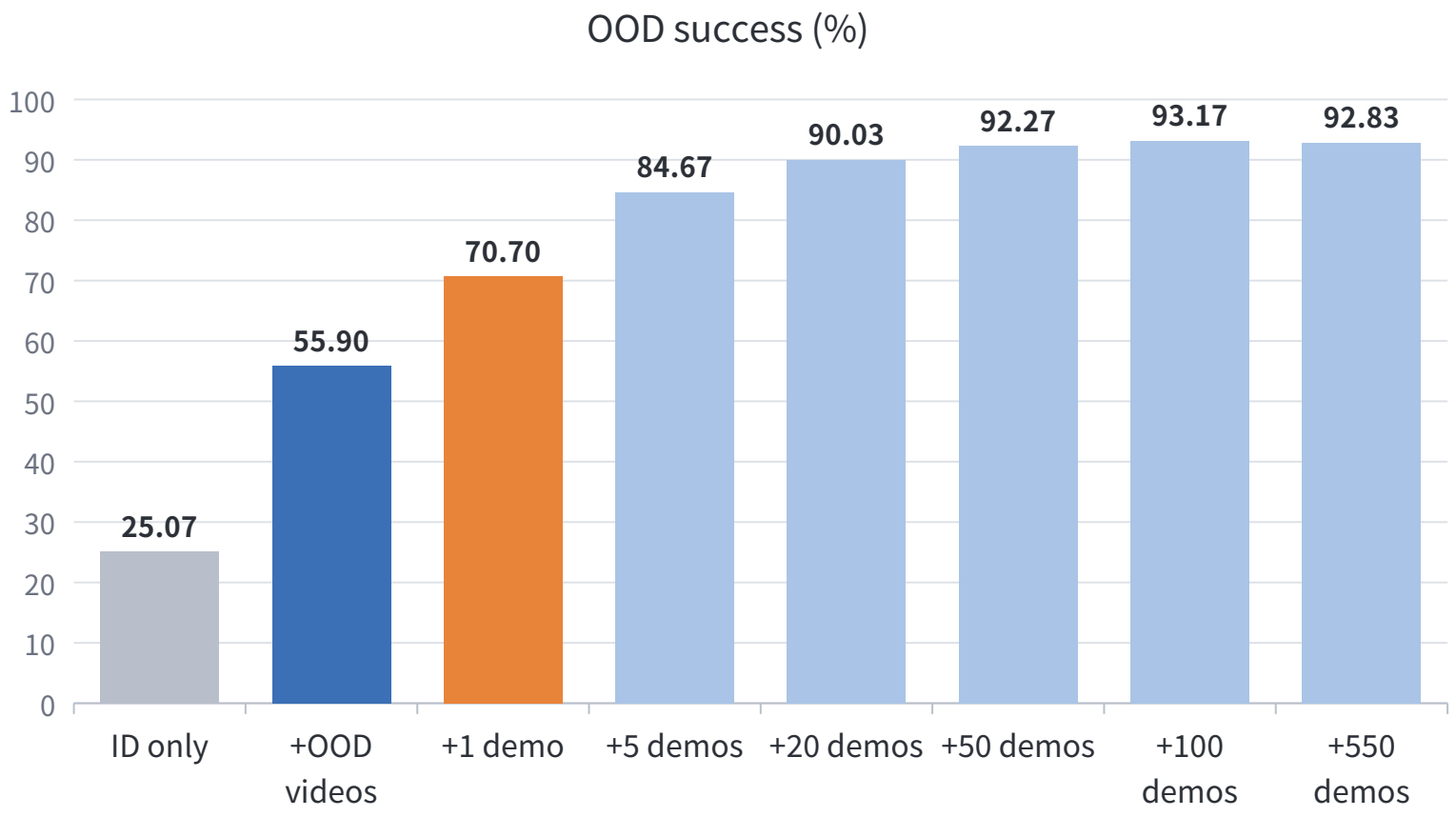





Source: arXiv 2610.40748 Fig. 2

Key Finding ①

# Unlabeled videos: OOD +30.8 pts; 1 demo per task: +14.8 more




X-axis: OOD supervision. From "+OOD videos" on, OOD videos are fixed at 550, then action demos per OOD task are added; ID data is always 550 videos + 550 actions.

**+30.83**

**Unlabeled OOD videos only**

ID stays 91.96%; overall 71.81→81.14

**3.6%**

**20 action demos per task**

OOD 90.03%, only 2.8 below full 92.83%

**≈Saturated**

**Beyond 100 demos**

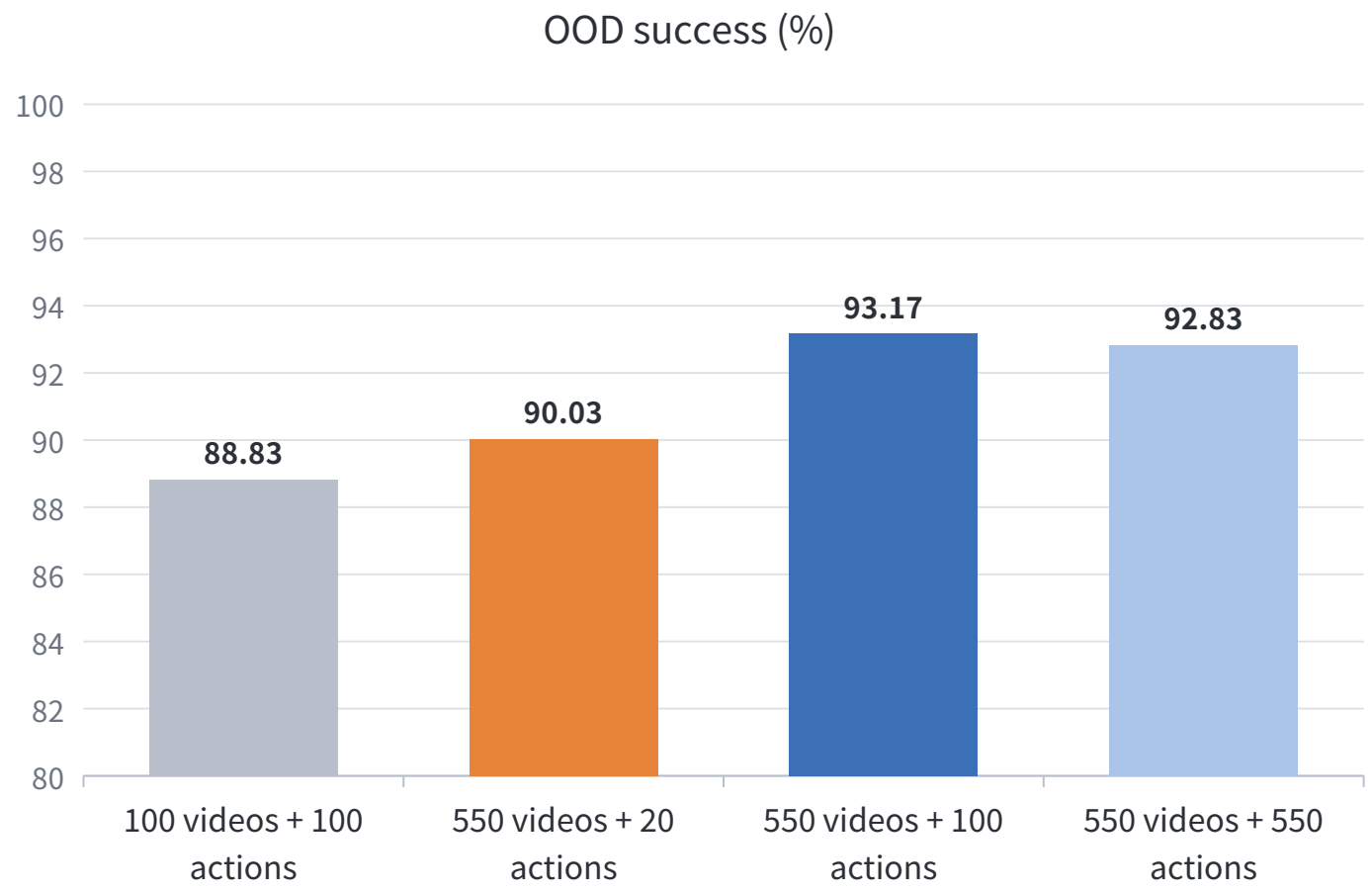
100: 93.17% vs 550: 92.83%



Source: arXiv 2610.40748 Table 2 (RoboTwin 2.0)

Key Finding ②

# Once demos suffice, scaling video beats adding action labels




Note: y-axis starts at 80% to magnify differences. Action counts are demos per OOD task.

**Videos 100 → 550 (actions = 100)**

OOD **88.83 → 93.17 （+4.34）**

**Actions 100 → 550 (videos = 550)**

OOD **93.17 → 92.83 (no gain)**

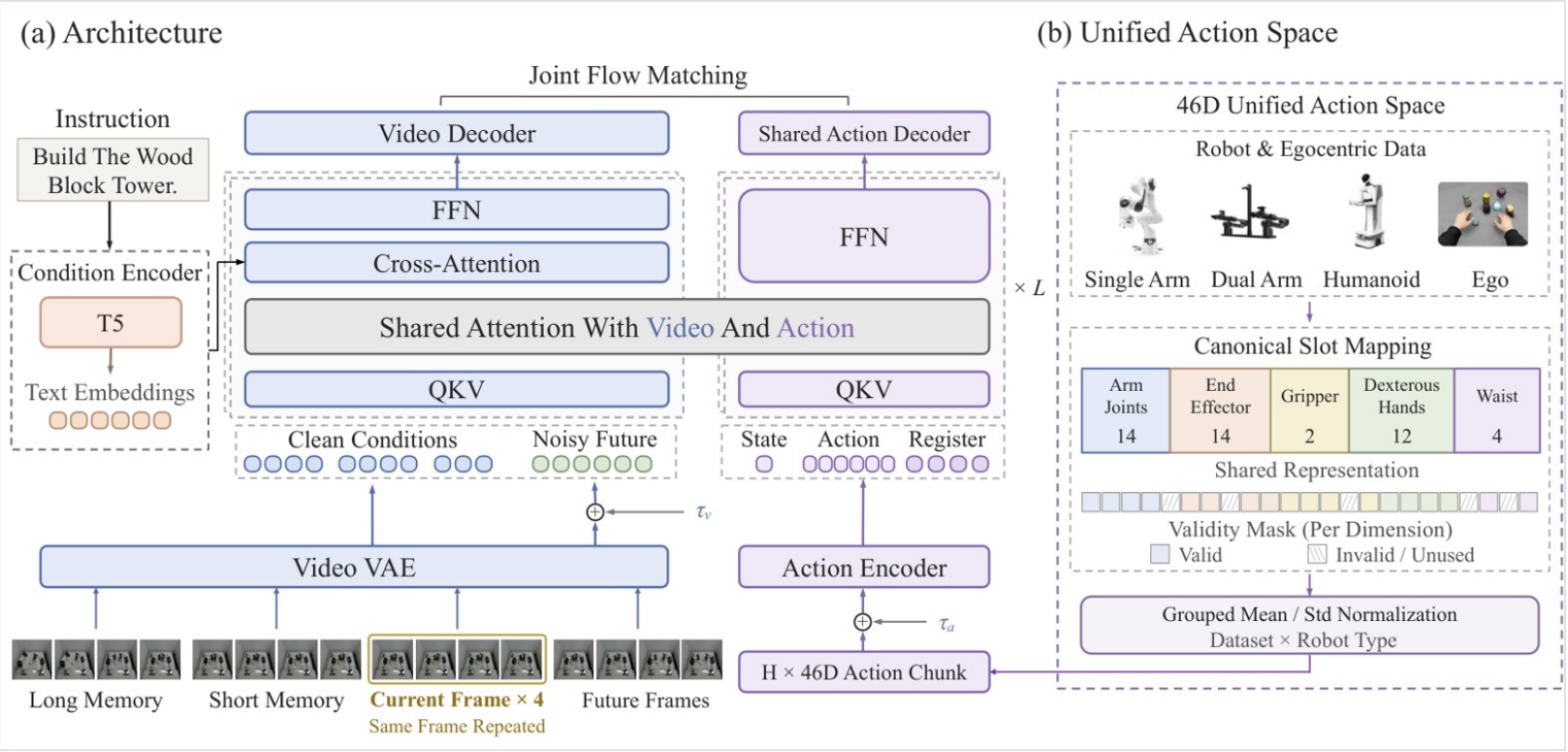
**1/5 the action labels does better**

550 videos + 20 actions **90.03** > 100 videos + 100 actions 88.83



Source: arXiv 2610.40748 Table 2 (Exp. 5/7/8/9)

# MoT: video & action experts share attention; joint flow matching




## Two decoupled experts

Separate QKV/FFN/decoders; shared joint self-attention links them each layer; video expert cross-attends to T5 instruction features.

## Joint flow matching

Video and action timesteps τv, τa sampled independently; linear noising, regressing velocity ε − x; equal video/action loss weights.

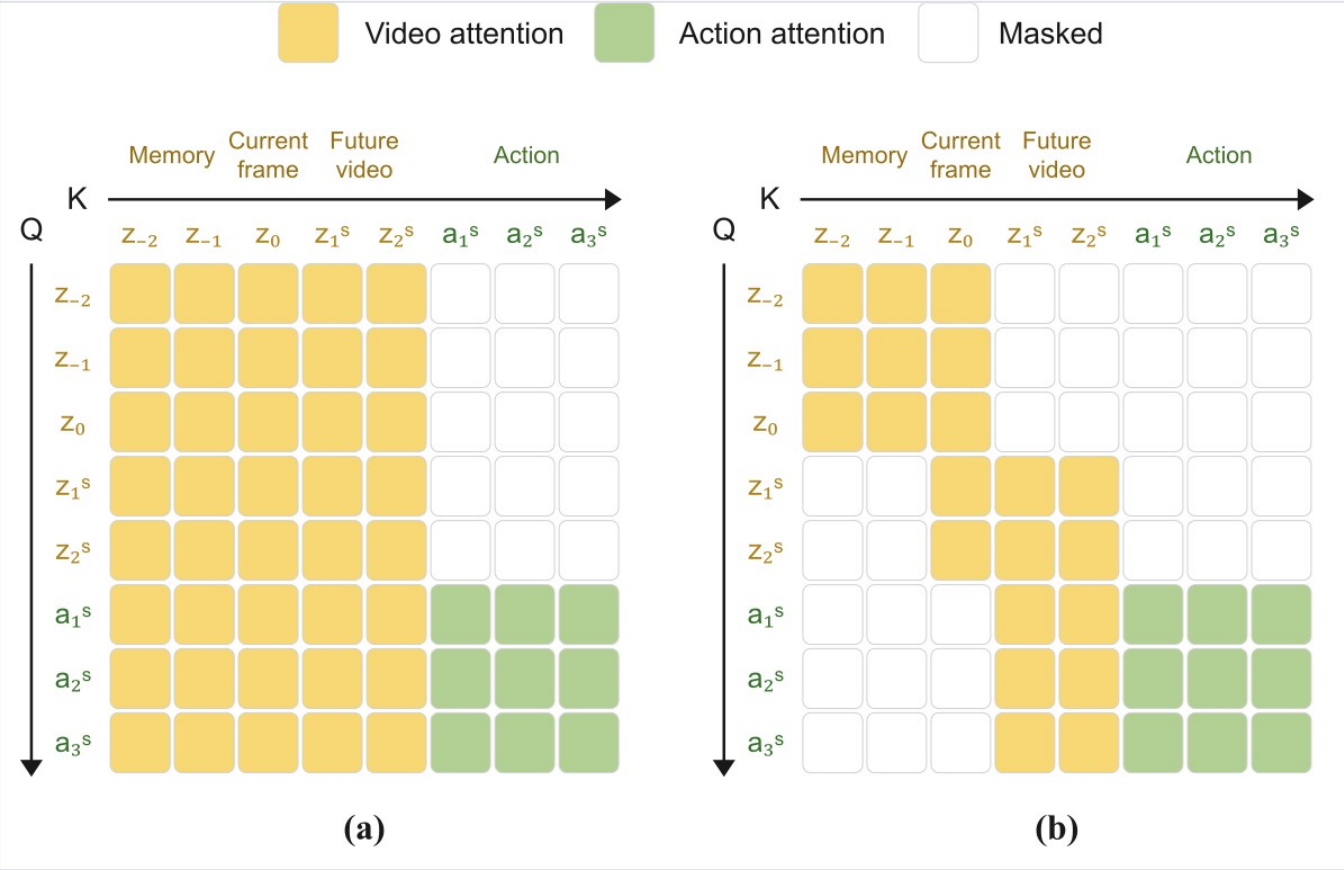
## Input structure

Video VAE encodes long-term / short-term memory / current frames ×4 / future frames; action side: H×46D action chunk + state token.



Source: arXiv 2610.40748 Fig. 3

# Post-training mask: memory/current → predicted future → action




## (a) Pre-training: dense video mask

Video attends only to video; generation is action-independent;

actions attend to all video + action tokens.

## (b) Post-training: directed pathways

Memory ↔ current frames; future video → current + future video;

**actions → future video + actions only**

Actions must rely on the "predicted visual evolution".

## Exceptions & side effects

Memory-heavy tasks (RMBench) use dense mask;

one-way mask lets inference cache video KV.



Source: arXiv 2610.40748 Fig. 4

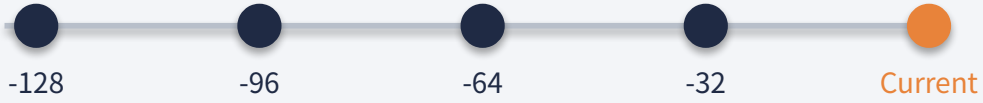
# Dual-scale memory adds context; Action Forcing ties actions to vision

## Dual-scale visual memory (4 frames each)

**Short-term: offsets [-8, -6, -4, -2], capturing local motion**




**Long-term: offsets [-128, -96, -64, -32]; stride = chunk size 32**




[zero pad, long ×4, short ×4, current ×4] → causal video VAE (4× temporal compression) → drop first latent

= 3 latents (long / short / current): **minimal history overhead**

Ablation: +short +0.67 / +long +1.22 (RoboTwin); RoboCasa +9.41 combined

## Action Forcing (post-training only)

**Training: independent noise level τa,k per group (g=4 shown)**



Darker = noisier; groups share τ, but noise vectors stay independent

**Inference: all actions share a timestep, jointly denoised**



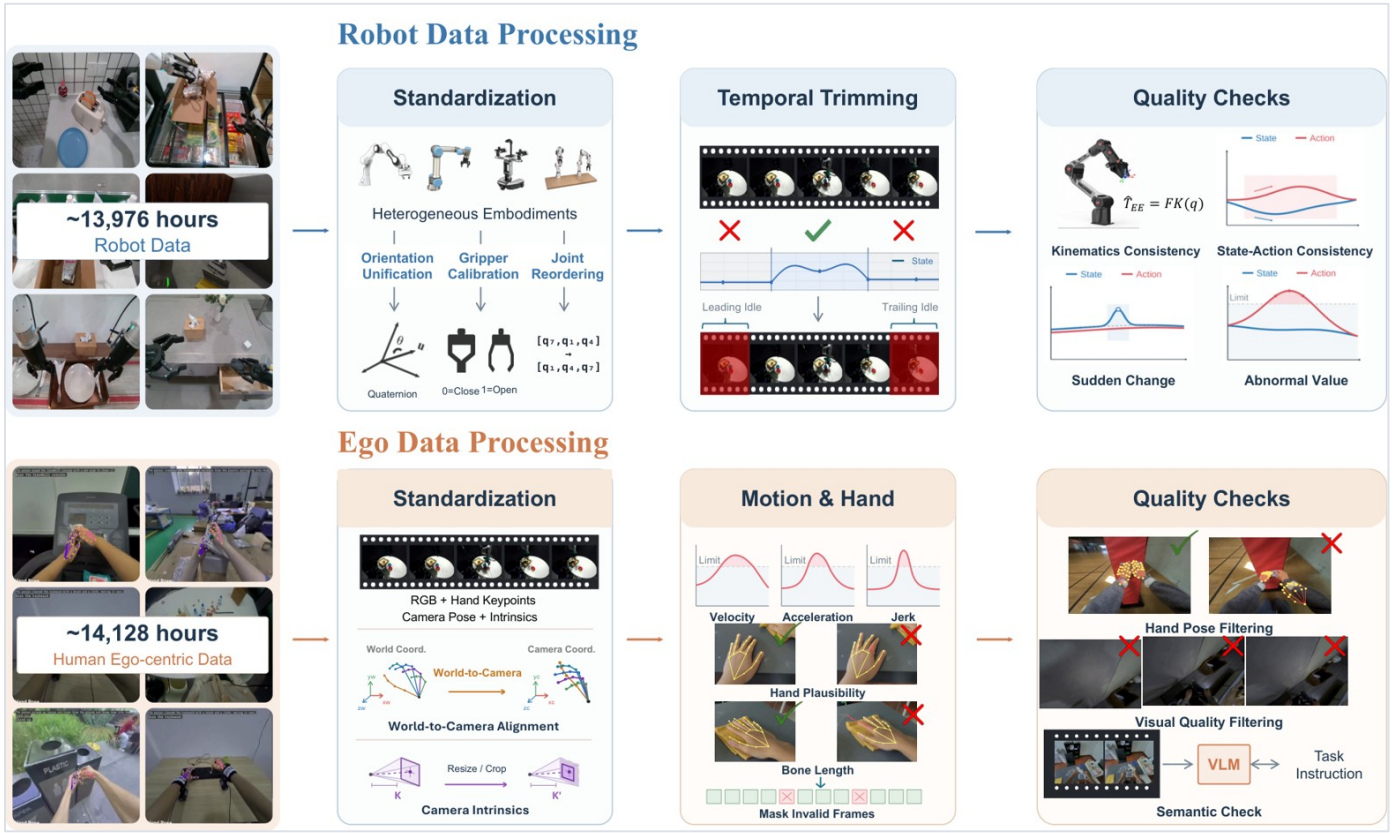
Motivation: action tokens at very different noise levels are hard to model in isolation → greater reliance on action context and future video features

Ablation: RoboTwin 91.28 → 93.31 (+2.03), RoboCasa +1.67, RoboDojo +2.02



Source: arXiv 2610.40748 § 2.1.2–2.1.3, § 2.3.3, Table 6 (schematic ours)

# 22k hours of robot + egocentric human data, one 46-D action space




**9,730 h** — Robot data (1.276M clips, 10 sources)

**12,592 h** — Egocentric human data (2.106M clips, 7 sources)

**22,321 h** — Total after filtering; 79.4% of duration retained

Pretrain: robot 0.7/human 0.3; 64 GPUs, bs 1536, 60k steps

## 46-D unified action space

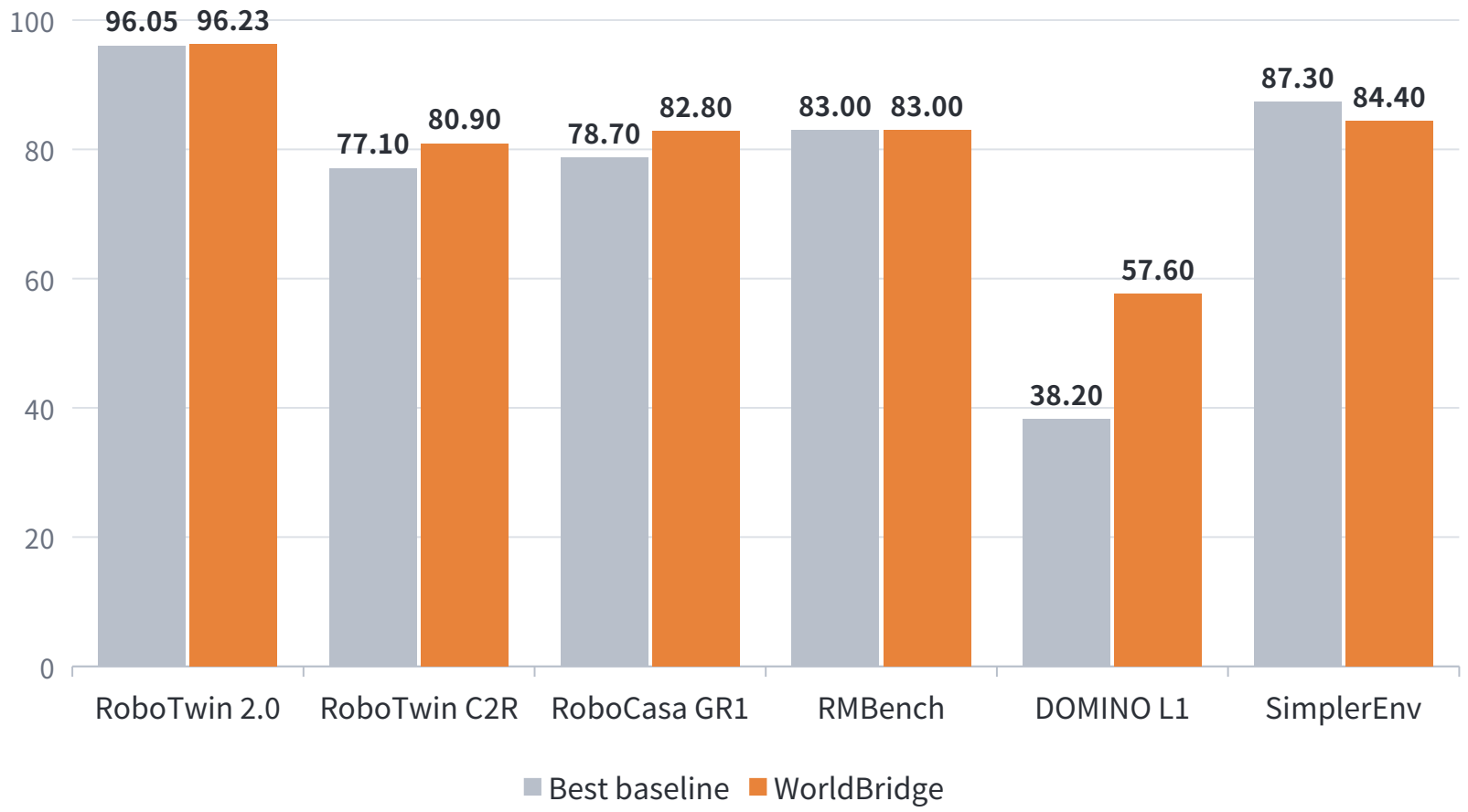
| Arm joints 14 | EEF pose 14 (rel. to current EEF) | | Dexterous hands 12 | Waist 4 |
|---|---|---|---|---|

Green = 2-D gripper; validity mask zeros & loss-masks missing dims; norm. stats per dataset × robot type



Source: arXiv 2610.40748 Fig. 5, Table 1, § 2.3.2

# Simulation: leads most; DOMINO w/o fine-tuning beats #2 by 19.4 pts




Best baselines: Kairos / Qwen-RobotManip-Context / Long-WAM / MemoryWAM / DynamicWAM / GalaxeaVLA (G0.5)

### DOMINO: static→dynamic

Reuses RoboTwin weights, no fine-tuning on dynamic tasks; SR 57.6 / MS 64.6

### C2R randomized: 73.4%

Fast-WAM 1.9%, AHA-WAM 3.2%: near-total failure here

### Not first everywhere

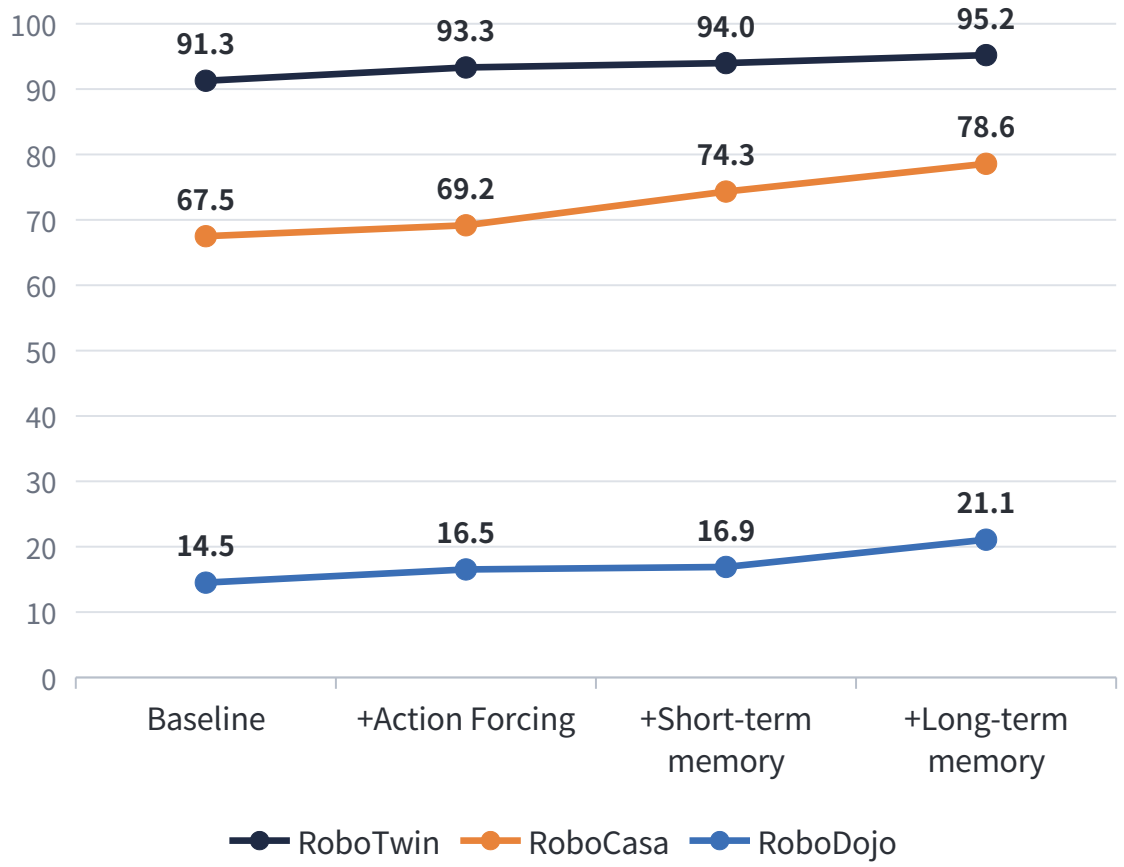
SimplerEnv 84.4 < 87.3; RoboDojo Score 26.03 best, but SR 19.15 < DM0.5 19.34



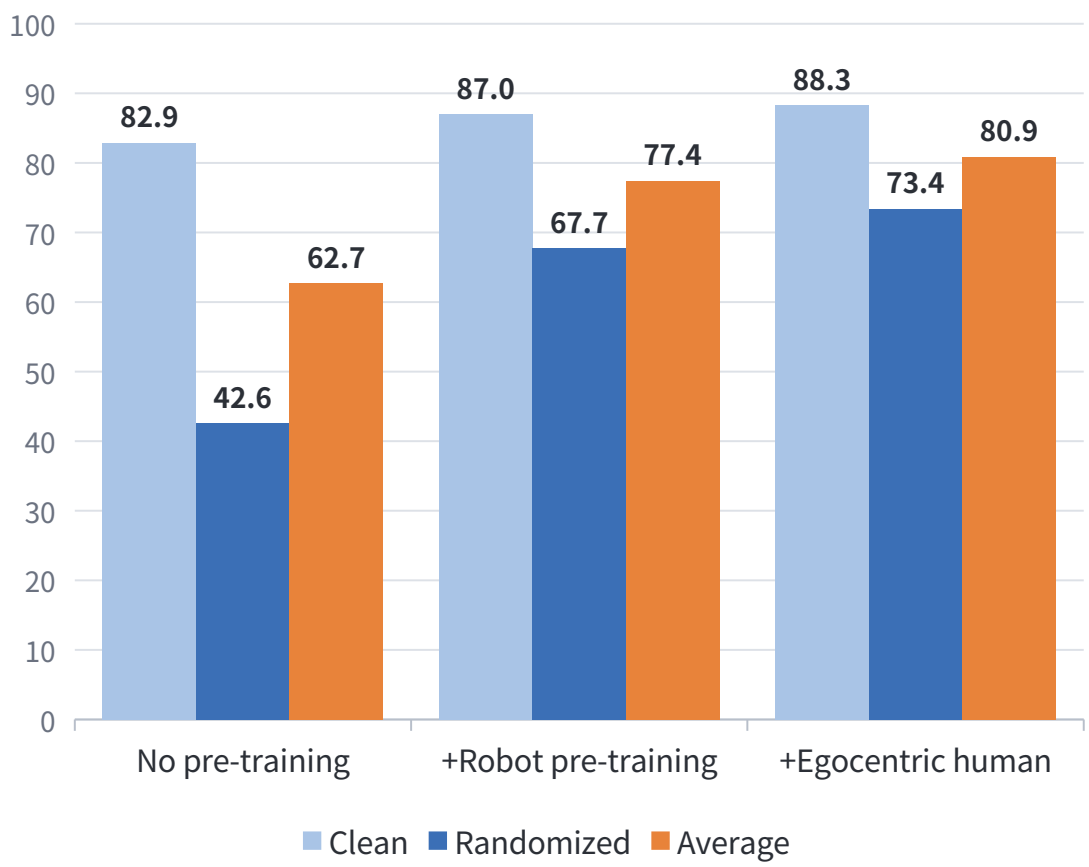
Source: arXiv 2610.40748 Tables 3/4/5

# Ablations: all parts help; pre-training matters most in random scenes

## Table 6 | Cumulative ablation (success %)



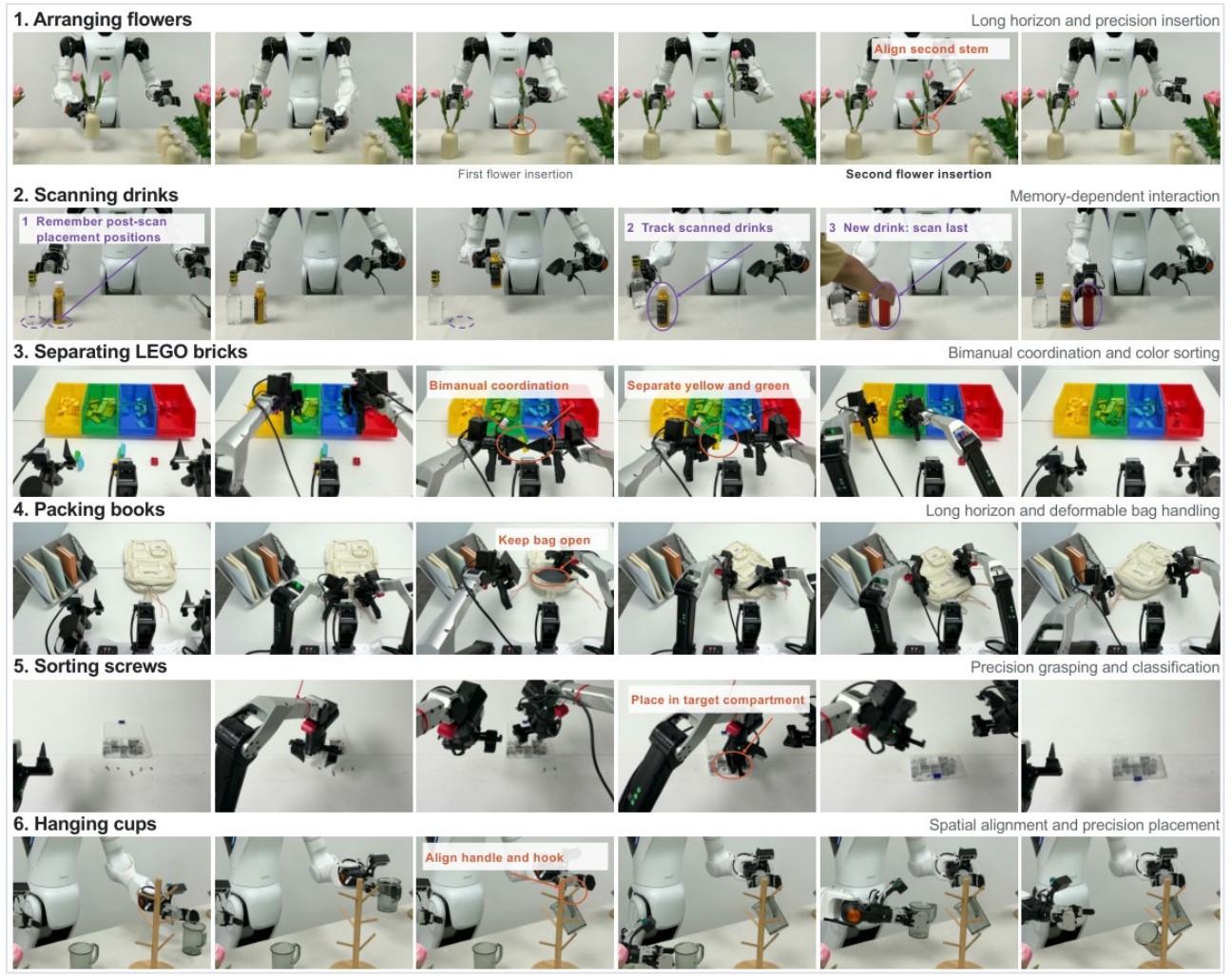

## Table 7 | Pre-training data (RoboTwin Clean→Random, %)






Source: arXiv 2610.40748 Table 6 / Table 7

# 2 real robots, 6 tasks: long-horizon, fine, memory, bimanual




## AgiBot G2

**Arrange flowers** long-horizon + precise insertion

**Scan drinks** memory: placement, scanned, new one last

**Hang cups** alignment, orientation-sensitive placing

## ARX AC-1

**Separate LEGO** bimanual coordination + color sorting

**Sort screws** fine grasping & sorting of small parts

**Pack books** long-horizon + deformable bag

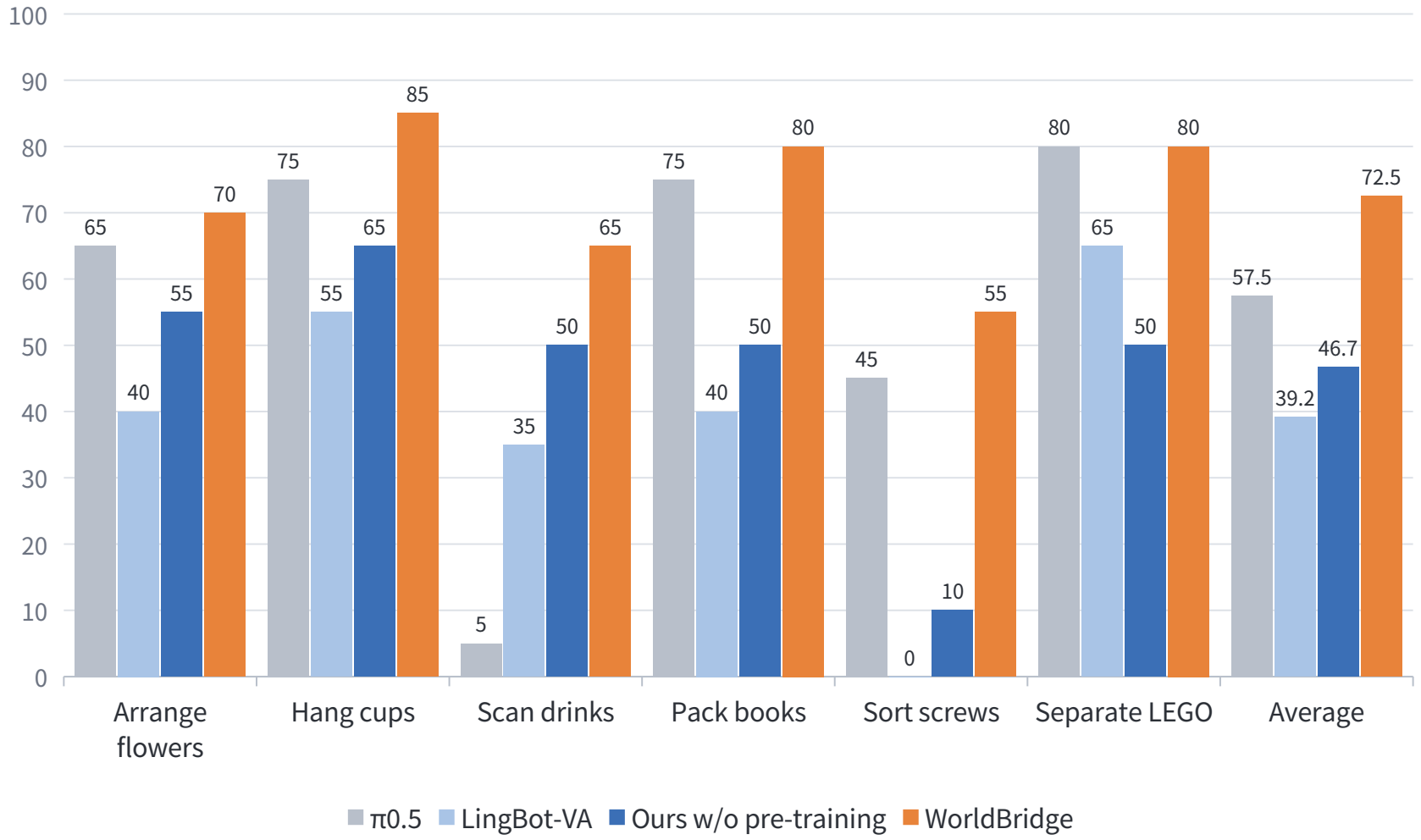
One pre-trained checkpoint fine-tuned per platform (G2 16-D / AC-1 14-D native commands, via adapters)

All methods fine-tuned on the same demos; 20 trials per task; success requires no human assistance



Source: arXiv 2610.40748 Fig. 6, § 3.3

# Real-robot avg. 72.5% (π0.5 57.5%); joint pre-training adds +25.8 pts




**65% vs 5%**

Memory task “Scan drinks”
WorldBridge vs π0.5

**+25.8**

Avg. gain from joint pre-training
( 46.7 → 72.5 )

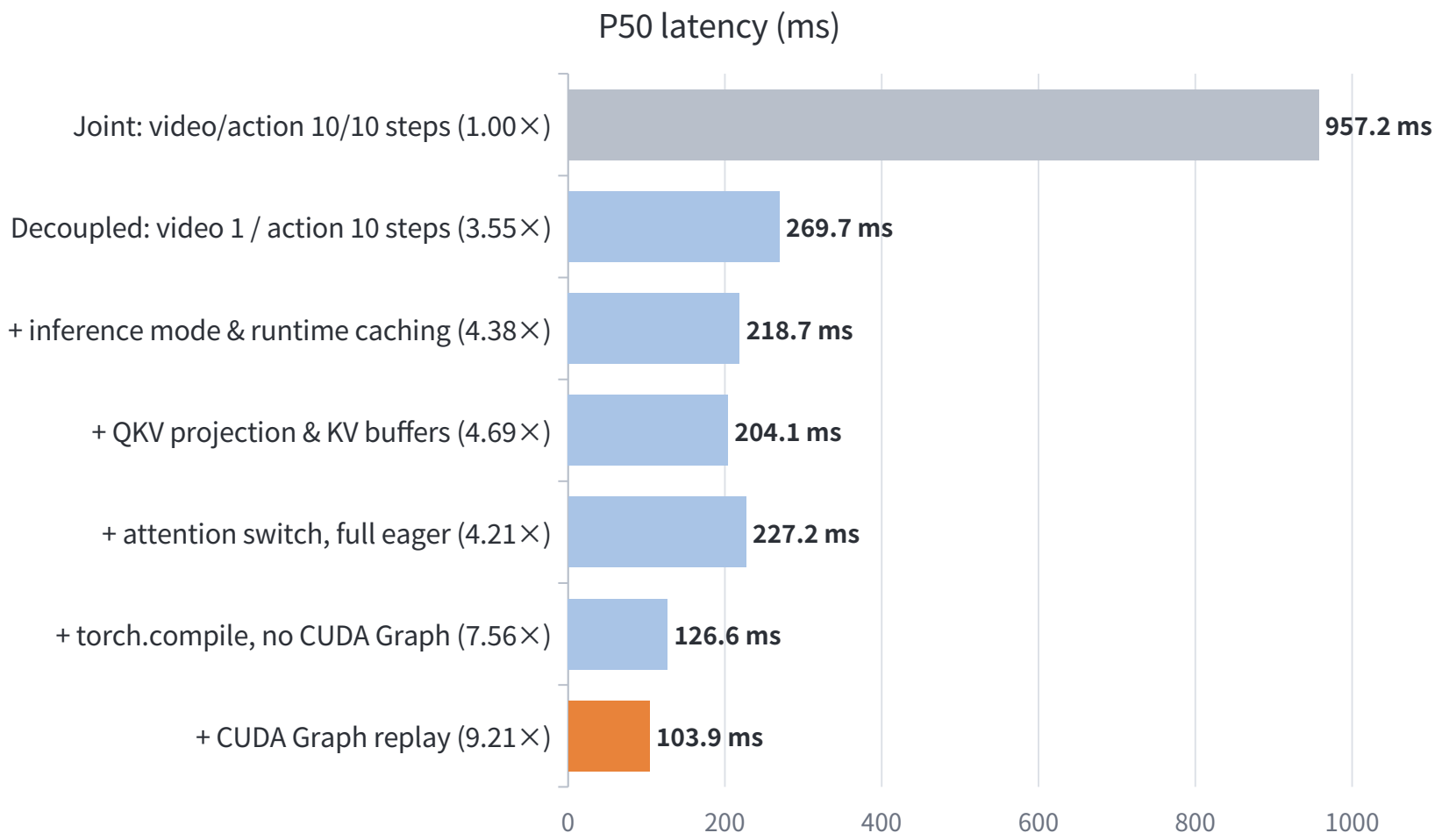
**10 → 55**

Sort screws: largest pre-training gain
(yet still the hardest task)



Source: arXiv 2610.40748 Fig. 7 (4 methods shown)

# Inference 9.2× faster: 957 ms → 104 ms per call, ~7 Hz on robot




### Key: decoupled denoising

Video denoised for 1 step only, caching per-layer KV; remaining steps run actions only; one-way mask makes reuse valid (though approximate)

### Engineering stack

Compilation + CUDA Graph: ~2× more; P95 105.6 ms

### Real-robot loop

Async execution + timestamp stitching; ~7 Hz incl. communication



Source: arXiv 2610.40748 Table 8, § 2.4

# Clear framing, solid engineering, but idealized generalization setup

## Strengths

1. Clean decomposition: video coverage vs. video→action bridge; controlled experiments quantify marginal gains
2. Architecture matches the thesis: post-training mask forces action ← predicted future
3. Thorough engineering: 46D unified action space, dual-pipeline data cleaning, 9.2× faster inference
4. Broad evaluation: 6 sim benchmarks (incl. memory, dynamic, real-to-sim) + 2 real robots

## Limitations & caveats

1. OOD "videos" are unlabeled demos from the same simulator and robot—far easier than human/web video; ID/OOD split unspecified
2. OOD video prediction quality is not measured separately, so failures can't be traced to video vs. bridge
3. SOTA claims need caveats: not first on SimplerEnv or RoboDojo SR; tied on RMBench; RoboTwin near saturation
4. Video gets only 1 denoising step at inference, in tension with the narrative; only 20 real-robot trials per task

Summary & Takeaways

# Task generalization = broader visual experience + a transferable “video→action bridge”

| | |
|---|---|
| **Data strategy** | Unlabeled videos widen coverage + few paired demos calibrate the bridge: 20/task reaches 90.03% (full data 92.83%) |
| **Model design** | Dual experts + directed attention masks + dual-scale memory + Action Forcing; 22k-hour mixed pre-training |
| **Next steps** | Validate with human / cross-embodiment videos; evaluate and strengthen the bridge (IDM) separately to cut new-task action labels |

arXiv 2610.40748 • Project page: worldbridge-jd.github.io